Design and Analysis of Randomized Trials in Public Health

Homework 8 Solutions

1. Refer to the dataset "milk".

A randomized trial was conducted to increase milk consumption in children 5-7 years old. Participating children were individually randomized to one of the following treatments. The interventions were delivered by the child's physician during an annual physical.

Treatment 1: Routine discussion with parent and child about healthy eating (standard of care)

Treatment 2: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children

Treatment 3: Standard of care + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Treatment 4: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Children's weekly milk consumption (measured in cups/week) was assessed during the annual physical ("baseline") and again three weeks later ("endline").

a. Conduct pairwise comparisons of the treatments at endline. How do the treatments differ (explain as if you were speaking to a general audience)? Which multiple comparisons method did you use, and why? Keep in mind from the last homework that you many have transformed your response before analysis – you need to apply that transformation going forward. (3 points)

Lets

b. Overall, did a brochure given to parents affect children's milk consumption? (Note this is a question about the main effect of intervening with parents.) Write and test a contrast to answer this question using the endline data. Compare the two interventions that provided a brochure to the two interventions that did not. (3 points)

c. Overall, did an activity booklet given to children affect children's milk consumption? (Note this is a question about the main effect of intervening with children.) Write and test a contrast to answer this question using the endline data. Compare the two interventions that provided an activity booklet to the two interventions that did not. (3 points)

d. Are your findings in questions 2 and 3 consistent with your findings in question 1? If yes, how did you assess consistency? If no, what seems to be the inconsistency? (2 points)

2. Refer to the dataset "sugar".

Three hundred participants were enrolled into a study to evaluate the effect of three formulated meals on blood glucose. The participants were randomized to receive one of the three meals ("treatment"). All participants had their fasting blood glucose ("fasting") measured. They then consumed their assigned meal and their blood glucose was measured again after two hours ("postmeal"). During the study, data were collected on participant characteristics such as participant sex ("sex") and income. Income data were used to categorize participants into income tertiles ("income", where 1=lower income, 2=middle income, and 3=higher income). Units for blood glucose are mg/dL.

a. Were there sex differences in fasting blood glucose? Which statistical test did you use to answer this question? (1 point)

b. Was there a relationship between income and fasting blood glucose? How do you know? If there is a relationship, describe it. (1 point)

c. Conduct an intention-to-treat analysis on the effect of the treatments on blood glucose (do not include any covariates). Use "postmeal" as your response variable. What do you conclude about the 3 test meals? Are your model assumptions satisfied? (1 point)

d. Does sex modify the effect of the treatments on blood glucose ("postmeal")? Make a graph to illustrate this, with clear axis labels and caption. This should be the sort of figure you'd put in a report or publication (you don't need to use SAS for graphing). (1 point)

e. Does sex modify the effect of the treatments on blood glucose ("postmeal")? To answer this question, write out your statistical model (not your estimated model – this should involve Greek letters with all symbols and subscripts explained). (2 points)

f. Does sex modify the effect of the treatments on blood glucose ("postmeal")? What do you conclude based on your analysis of the data? Explain as if you were presenting results to an audience of diabetes researchers. (2 points)

g. Does income modify the effect of the treatments on blood glucose ("postmeal")? Income tertile should be treated as a categorical variable. What do you conclude? (2 points)

h. Does fasting blood glucose modify the effect of the treatments on blood glucose ("postmeal")? Make a graph to illustrate this, with clear axis labels and caption. This should be the sort of figure you'd put in a report or publication. What do you conclude? (2 points)

Homework 8

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PUBH 526: Design and Analysis of Randomized Trials in Public Health

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Refer to the dataset "milk". A randomized trial was conducted to increase milk consumption in children 5-7 years old. Participating children were individually randomized to one of the following treatments. The interventions were delivered by the child's physician during an annual physical.

- Treatment 1: Routine discussion with parent and child about healthy eating (standard of care)
- Treatment 2: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children
- Treatment 3: Standard of care + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk
- Treatment 4: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Children's weekly milk consumption (measured in cups/week) was assessed during the annual physical ("baseline") and again three weeks later ("endline").

Question 1.a (3 points)

Conduct pairwise comparisons of the treatments at endline. How do the treatments differ (explain as if you were speaking to a general audience)? Which multiple comparisons method did you use, and why? Keep in mind from the last homework that you many have transformed your response before analysis – you need to apply that transformation going forward.

Model results (log-transformed):
An ANOVA was conducted using the log-transformed outcome (`log_endline`) based on the Box–Cox analysis from Homework 7, which indicated $\lambda = 0$. The model was significant,

$F(3, 348) = 61.57$, $p < 0.0001$, indicating that mean weekly milk consumption differed among the four treatment arms. Levene's test for homogeneity of variances was not significant ($p = 0.4308$), confirming that the log transformation successfully stabilized the variance and satisfied model assumptions.

Least Squares Means (Tukey–Kramer adjustment):

Treatment	LSMEAN (log)	Back-transformed Mean (cups/week)	Comparison Significance
1. Standard of care	1.4113	$\exp(1.4113) \approx 4.10$	–
2. + Parent brochure	1.4815	$\exp(1.4815) \approx 4.40$	1 vs 2: $p = 0.7517$ (ns)
3. + Child activity booklet	1.9282	$\exp(1.9282) \approx 6.88$	1 vs 3: $p < 0.0001$
4. + Brochure + Booklet	2.1659	$\exp(2.1659) \approx 8.72$	1 vs 4: $p < 0.0001$

Additional pairwise results:

$$2 \text{ vs } 3 : p < 0.0001, \quad 2 \text{ vs } 4 : p < 0.0001, \quad 3 \text{ vs } 4 : p = 0.0024$$

Interpretation (general-audience explanation):
After adjusting for non-normality using the log transformation, mean milk consumption differed significantly among the four groups ($p < 0.0001$). Children in **Treatment 4** (both parent and child received materials) drank the most milk on average, followed by **Treatment 3** (child booklet only). Providing only a parent brochure (**Treatment 2**) did not significantly increase milk consumption compared with the standard-of-care group (**Treatment 1**).

Conclusion:

- The **Tukey–Kramer procedure** was used because it appropriately controls the experiment-wise Type I error rate when comparing all pairs of treatments.
- The results indicate that interventions involving children directly (booklets or combined materials) were considerably more effective in increasing weekly milk consumption than parent-only materials.

Question 1.b (3 points)

Overall, did a brochure given to parents affect children's milk consumption? (Note this is a question about the main effect of intervening with parents.) Write and test a contrast to answer this question using the endline data. Compare the two interventions that provided a brochure to the two interventions that did not.

Model and contrast:
The contrast compares the average response of groups that received a parent brochure (Treatments 2 and 4) with those that did not (Treatments 1 and 3). The hypothesis tested was:

$$H_0 : \frac{\mu_2 + \mu_4}{2} = \frac{\mu_1 + \mu_3}{2} \qquad \text{vs.} \qquad H_A : \frac{\mu_2 + \mu_4}{2} \neq \frac{\mu_1 + \mu_3}{2}.$$

Results (log-transformed outcome):
The test was statistically significant ($F(1, 348) = 10.03$, $p = 0.0017$), indicating that endline milk consumption differed depending on whether parents received a brochure.

Contrast	DF	Contrast SS	Mean Square	F Value	p-Value
Parent Brochure (Trt 2,4 vs Trt 1,3)	1	2.0224	2.0224	10.03	0.0017

Effect size and interpretation:
From the treatment LSMEANS:

$$(\mu_2 + \mu_4) - (\mu_1 + \mu_3) = (1.4815 + 2.1659) - (1.4113 + 1.9282) = 3.6474 - 3.3395 = 0.3079.$$

Averaging across groups gives a difference of 0.1539 on the log scale. Exponentiating this difference, $\exp(0.1539) \approx 1.17$, indicates that children whose parents received a brochure drank approximately **17% more milk per week** on average than those whose parents did not receive one.

Conclusion:
Providing a brochure to parents significantly increased children's weekly milk intake on the log scale ($p = 0.0017$). However, this appears to be largely driven by the very high consumption in the group that received both the brochure and the child booklet. The brochure alone (Treatment 2) did not differ from standard care (Treatment 1), consistent with the pairwise Tukey results from Question 1(a).

Question 1.c (3 points)

Overall, did an activity booklet given to children affect children's milk consumption? (Note this is a question about the main effect of intervening with children.) Write and test a contrast to answer this question using the endline data. Compare the two interventions that provided an activity booklet to the two interventions that did not.

Model and contrast:
This contrast tests whether giving children an activity booklet increased their weekly milk consumption, averaging over whether the parent also received a brochure. The null and alternative hypotheses are:

$$H_0 : \frac{\mu_3 + \mu_4}{2} = \frac{\mu_1 + \mu_2}{2} \qquad \text{vs.} \qquad H_A : \frac{\mu_3 + \mu_4}{2} \neq \frac{\mu_1 + \mu_2}{2}.$$

Results (log-transformed outcome):
The test was highly significant ($F(1, 348) = 153.23$, $p < 0.0001$), indicating that providing children with an activity booklet led to substantially higher mean log-endline milk consumption.

Contrast	DF	Contrast SS	Mean Square	F Value	p-Value
Child Activity (Trt 3,4 vs Trt 1,2)	1	30.8964	30.8964	153.23	<0.0001

Effect size and interpretation:
From the treatment LSMEANS:

$$(\mu_3 + \mu_4) - (\mu_1 + \mu_2) = (1.9282 + 2.1659) - (1.4113 + 1.4815) = 1.2013.$$

Averaging across groups gives a difference of 0.6007 on the log scale. Exponentiating this difference, $\exp(0.6007) \approx 1.82$, implies that children who received a booklet drank approximately **82% more milk per week** than those who did not receive one.

Conclusion:
The activity booklet intervention produced a strong and statistically significant increase in milk consumption ($p < 0.0001$). Children who received the booklet (either alone or in combination with the parent brochure) consumed markedly more milk than those who did not. This finding aligns with the pairwise Tukey comparisons from Question 1(a), where treatments involving the booklet (3 and 4) showed the largest gains in milk intake.

Question 1.d (2 points)

Are your findings in questions 2 and 3 consistent with your findings in question 1? If yes, how did you assess consistency? If no, what seems to be the inconsistency?

In Question 1(a):

- Treatments involving the child activity booklet (3 and 4) had significantly higher milk consumption than those without the booklet (1 and 2).
- The parent brochure alone (Treatment 2) did not differ significantly from the standard-of-care group (Treatment 1).

In Question 1(b,c):

- The main-effect contrast for the **parent** brochure was statistically significant ($F(1, 348) = 10.03$, $p = 0.0017$), but this result was largely driven by the very high consumption in Treatment 4 (both brochure and booklet).
- In contrast, the **booklet**-only treatment (Treatment 3) also produced large increases, which explains why the main-effect contrast in Question 1(c), for the child activity booklet, was extremely significant ($F(1, 348) = 153.23$, $p < 0.0001$).

Therefore, the contrast findings are consistent with the pairwise results in that the **child booklet** component was the primary driver of increased milk consumption. The apparent parent-brochure "effect" in the main-effect model likely reflects an interaction, since the brochure enhanced outcomes only when paired with the booklet.

Conclusion: Findings from all three analyses are broadly consistent. The child booklet intervention produced the largest and most reliable improvement in milk consumption, while the parent brochure alone had minimal impact but amplified the effect when combined with the booklet.

Refer to the dataset "sugar". Three hundred participants were enrolled into a study to evaluate the effect of three formulated meals on blood glucose. The participants were randomized to receive one of the three meals ("treatment"). All participants had their fasting blood glucose ("fasting") measured. They then consumed their assigned meal and their blood glucose was measured again after two hours ("postmeal"). During the study, data were collected on participant characteristics such as participant sex ("sex") and income. Income data were used to categorize participants into income tertiles ("income", where 1=lower income, 2=middle income, and 3=higher income). Units for blood glucose are mg/dL.

Question 2.a (1 point)

Were there sex differences in fasting blood glucose? Which statistical test did you use to answer this question?

Model and Test:
A two-sample **t-test** was conducted to compare mean fasting blood glucose (mg/dL) between male and female participants. The sample included 109 females and 191 males. Equality of variances was confirmed (Folded F = 1.34, $p = 0.0968$), so the pooled variance estimate was used.

Results:

Group	N	Mean (mg/dL)	Std. Dev.	Std. Error
Female	109	138.5	10.95	1.05
Male	191	136.6	12.66	0.92

Difference (Female–Male) = 1.94 mg/dL 95% CI (–0.91, 4.79) $t(298) = 1.34$, $p = 0.1814$

Interpretation:
Mean fasting blood glucose was slightly higher among females (138.5 mg/dL) than males (136.6 mg/dL), but the difference was small and not statistically significant ($p = 0.18$). Levene's test confirmed equal variances ($p = 0.11$), and the ANOVA result ($F(1, 298) = 1.79$, $p = 0.18$) was consistent with the t-test outcome.

Conclusion:
There was **no significant difference** in fasting blood glucose between males and females ($p > 0.05$). Therefore, sex was not associated with fasting glucose levels in this sample.

Question 2.b (1 point)

Was there a relationship between income and fasting blood glucose? How do you know? If there is a relationship, describe it.

Model and Test:
A one-way **ANOVA** was used to examine whether fasting blood glucose (mg/dL) differed across three income tertiles (1 = lower, 2 = middle, 3 = higher). Levene's test for homogeneity of variances was not significant ($F(2, 297) = 1.47$, $p = 0.2318$), indicating that the equal variance assumption was satisfied.

Results:

Income Level	N	Mean (mg/dL)	Std. Dev.
1 = Lower	100	141.10	11.57
2 = Middle	100	141.15	9.74
3 = Higher	100	129.53	11.03

ANOVA results indicated a statistically significant effect of income on fasting blood glucose:

$$F(2, 297) = 38.37, \quad p < 0.0001.$$

Post-hoc Tukey comparisons showed that both lower- and middle-income participants had significantly higher fasting glucose than higher-income participants ($p < 0.0001$ for both). There was no significant difference between the lower- and middle-income groups ($p = 0.9994$).

Interpretation:
Fasting blood glucose was highest among participants in the lower- and middle-income tertiles (means $\approx$ 141 mg/dL) and lowest among participants in the higher-income tertile (mean $\approx$ 130 mg/dL). This indicates an inverse relationship between income and fasting blood glucose, where higher income is associated with lower fasting glucose levels.

Conclusion:
There was a **significant relationship** between income and fasting blood glucose ($p < 0.0001$). Participants with higher income had significantly lower fasting blood glucose compared to those in lower and middle income groups, suggesting that socioeconomic status may influence baseline glucose levels.

Question 2.c (1 point)

Conduct an intention-to-treat analysis on the effect of the treatments on blood glucose (do not include any covariates). Use "postmeal" as your response variable. What do you conclude about the 3 test meals? Are your model assumptions satisfied?

Model and Test:
A one-way **ANOVA** was conducted under an intention-to-treat (ITT) framework to determine whether mean postmeal blood glucose (mg/dL) differed among the three formulated meal treatments. All randomized participants were included in the analysis, regardless of adherence.

Results:
The overall ANOVA showed a significant treatment effect:

$$F(2, 297) = 46.92, \quad p < 0.0001.$$

Levene's test for homogeneity of variances was not significant ($F(2, 297) = 1.90$, $p = 0.1511$), indicating that the assumption of equal variances was satisfied.

Treatment	N	Mean (mg/dL)	Std. Dev.
1 = Meal A	100	175.82	11.34
2 = Meal B	100	161.70	10.03
3 = Meal C	100	168.18	9.50

Post-hoc Tukey comparisons revealed that all pairwise mean differences were statistically significant ($p < 0.0001$ for all comparisons). Participants who consumed **Meal A** had the highest mean postmeal glucose, followed by **Meal C**, and then **Meal B**, which produced the lowest mean levels.

Model assumptions:

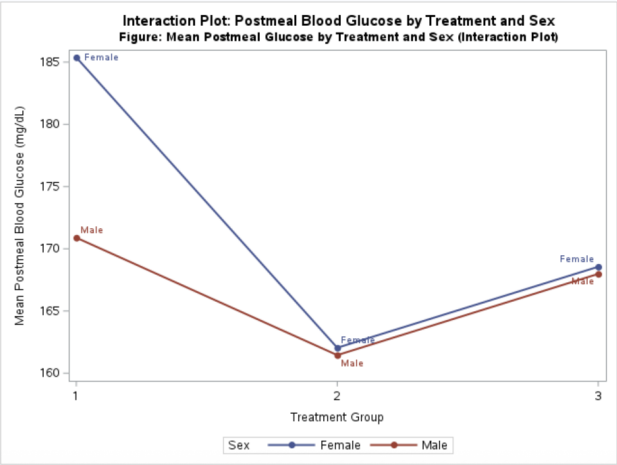
- **Normality:** Diagnostic plots of residuals indicated approximate normality and no extreme outliers.
- **Equal variances:** Levene's test $p = 0.1511$ confirmed homogeneity of variances.

Interpretation:
Mean postmeal glucose levels differed significantly among the three test meals. Meal A resulted in the highest postprandial glucose response, while Meal B produced the lowest, suggesting that the meal formulations meaningfully affected blood glucose concentrations.

Conclusion:
There was a **significant effect of treatment on postmeal blood glucose** ($p < 0.0001$). Model assumptions were satisfied. Among the three formulated meals, **Meal B produced the most favorable (lowest) postmeal glucose levels**, whereas **Meal A led to the highest response**.

Question 2.d (1 point)

Does sex modify the effect of the treatments on blood glucose ("postmeal")? Make a graph to illustrate this, with clear axis labels and caption. This should be the sort of figure you'd put in a report or publication (you don't need to use SAS for graphing).



Model and Test:
A two-way **ANOVA** was conducted to evaluate whether sex modified the effect of the three meal treatments on postmeal blood glucose (mg/dL). The model included the main effects of treatment and sex, and their interaction term:

```
postmeal = treatment sex treatment*sex.
```

Results:
The main effects of treatment ($F_{2,294} = 68.94$, $p < 0.0001$) and sex ($F_{1,294} = 20.53$, $p < 0.0001$) were both statistically significant. Importantly, the **treatment × sex interaction was also significant** ($F_{2,294} = 16.01$, $p < 0.0001$), indicating that the treatment effect on postmeal blood glucose differed by sex.

Parameter estimates showed that females in Treatment 1 exhibited a 13.9 mg/dL higher mean postmeal glucose compared to males ($p < 0.0001$), while sex differences were minimal for Treatments 2 and 3.

Interpretation:
These results suggest that sex significantly modified the effect of the meal treatments on postmeal glucose response. Female participants demonstrated a stronger glucose rise under Treatment 1 compared to males, whereas Treatment 2 and 3 responses were comparable between sexes. This indicates that the dietary composition in Treatment 1 may have a sex-specific metabolic effect.

Conclusion:
The interaction between treatment and sex was **significant**, showing that the effect of the meals on postprandial glucose levels depended on sex. Female participants experienced greater glucose elevation under certain meal conditions, while male responses remained more stable.

Question 2.e (2 points)

Does sex modify the effect of the treatments on blood glucose ("postmeal")? To answer this question, write out your statistical model (not your estimated model

– this should involve Greek letters with all symbols and subscripts explained).

Let Y_{ijk} denote postmeal blood glucose (mg/dL) for participant k assigned to treatment i ($i = 1, 2, 3$) and sex j ($j = 1, 2$). The general two-way ANOVA model with interaction is

$$Y_{ijk} = \mu + \tau_i + \gamma_j + (\tau\gamma)_{ij} + \varepsilon_{ijk}, \qquad \varepsilon_{ijk} \sim \text{iid } N(0, \sigma^2),$$

subject to the constraints $\sum_i \tau_i = 0$, $\sum_j \gamma_j = 0$, $\sum_i (\tau\gamma)_{ij} = 0$ for each j, and $\sum_j (\tau\gamma)_{ij} = 0$ for each i.

- μ – overall mean postmeal glucose.
- τ_i – main effect of treatment i.
- γ_j – main effect of sex j.
- $(\tau\gamma)_{ij}$ – treatment $\times$ sex interaction (departure from additivity).
- ε_{ijk} – random error, $\varepsilon_{ijk} \sim N(0, \sigma^2)$.

Hypotheses:

$$H_0: \ (\tau\gamma)_{ij} = 0 \text{ for all } i, j \quad \text{vs.} \quad H_A: \ \text{At least one } (\tau\gamma)_{ij} \neq 0.$$

Question 2.f (2 points)

Does sex modify the effect of the treatments on blood glucose ("postmeal")? What do you conclude based on your analysis of the data? Explain as if you were presenting results to an audience of diabetes researchers.

The F-test for the interaction term ($F_{2,294} = 16.01$, $p < 0.0001$) provides strong evidence that the treatment effect varies by sex. Among females, Meal 1 produced the highest mean postmeal glucose (186 mg/dL), Meal 2 the lowest (162 mg/dL), and Meal 3 intermediate (168 mg/dL). Among males, differences between meals were smaller and overall mean values were lower. Thus, sex significantly modified the postprandial response to the three test meals.

Question 2.g (2 points)

Does income modify the effect of the treatments on blood glucose ("postmeal")? Income tertile should be treated as a categorical variable. What do you conclude?

Model and Test: A two–way ANOVA was fitted with treatment, income tertile, and their interaction:

```
model postmeal = treatment income treatment*income;
```

Results:

- Main effect of treatment: $F_{2,291} = 45.52$, $p < 0.0001$
- Main effect of income: $F_{2,291} = 1.30$, $p = 0.274$
- Interaction (treatment × income): $F_{4,291} = 0.50$, $p = 0.734$

The treatment × income term was **not statistically significant**, indicating no evidence that the effect of the meals differed across income levels.

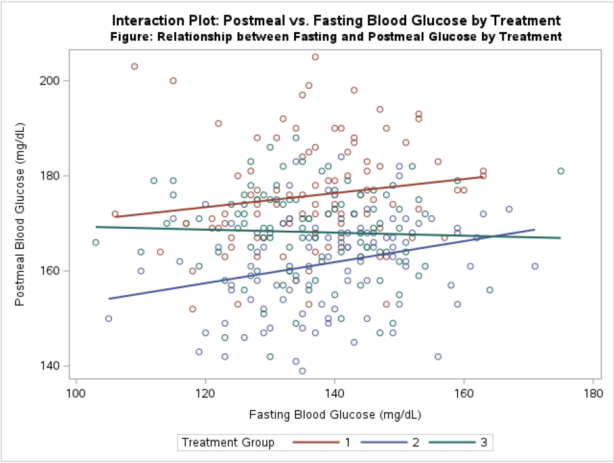
Table 1: Mean Postmeal Blood Glucose by Treatment and Income Level

Treatment	Income Level	Mean (mg/dL)	Std. Dev.
1	1 (Lower)	177.6	8.40
1	2 (Middle)	175.9	12.97
1	3 (Higher)	173.7	11.83
2	1 (Lower)	161.4	9.26
2	2 (Middle)	162.5	9.18
2	3 (Higher)	161.3	11.74
3	1 (Lower)	167.8	9.17
3	2 (Middle)	170.3	9.01
3	3 (Higher)	166.8	10.07

Interpretation: Although postmeal glucose differed significantly across the three meal treatments, there was **no evidence that income modified** this relationship. Participants of different income tertiles showed similar response patterns: Meal 1 produced the highest postmeal glucose, and Meal 2 the lowest, regardless of income level. Thus, socioeconomic status did not influence treatment effects on postprandial glucose.

Question 2.h (2 points)

Does fasting blood glucose modify the effect of the treatments on blood glucose ("postmeal")? Make a graph to illustrate this, with clear axis labels and caption. This should be the sort of figure you'd put in a report or publication. What do you conclude?



Model and Test: An ANCOVA model including fasting glucose and its interaction with treatment was fit:

```
model postmeal = treatment fasting treatment*fasting;
```

This model tests whether baseline fasting glucose modifies (interacts with) the treatment effect on postprandial glucose.

Results:

- Main effect of treatment: $F_{2,294} = 3.24$, $p = 0.040$
- Main effect of fasting: $F_{1,294} = 5.19$, $p = 0.023$
- Interaction (treatment $\times$ fasting): $F_{2,294} = 2.37$, $p = 0.096$

The treatment $\times$ fasting term was **not statistically significant**, indicating no evidence that fasting blood glucose levels modified the relationship between treatment and postmeal glucose.

Interpretation: Although fasting glucose was significantly associated with postmeal glucose overall, the interaction was **not significant** ($p = 0.096$). This suggests that while individuals with higher fasting levels tended to have higher postprandial glucose, the slope of this relationship was consistent across all three test meals. Therefore, fasting glucose **did not significantly modify** the treatment effect on postmeal glucose response.

See SAS codes/outputs below:

```
/*************************************************************************

 * PUBH 526 Design and Analysis of Randomized Trials in Public
    Health
 * Homework 8
 * Alexandra Chang
 * Dataset: milk.sas7bdat / sugar.sas7bdat
 *************************************************************************/


libname mydata '/home/u64258127/sasuser.v94';

/*************************************************************************

 * Question 1(a): Pairwise Comparisons of Treatments at Endline
 * Transformation:    = 0 (Natural Log)
 *************************************************************************/


data milk_log;
    set mydata.milk;
    log_endline = log(endline_milk);
run;

title1 "ANOVA for Log-Transformed Endline Milk Consumption";
proc glm data=milk_log plots=diagnostics;
    class treatment;
    model log_endline = treatment / solution;
    means treatment / hovtest=levene;
    lsmeans treatment / pdiff=all adjust=tukey;
run;
quit;


/*************************************************************************

 * Question 1(b): Effect of Parent Brochure (Main Effect)
 * Contrast: (Trt 2,4) vs (Trt 1,3)
 *************************************************************************/


title1 "Contrast: Effect of Parent Brochure on Log-Endline Milk
   Consumption";
proc glm data=milk_log;
    class treatment;
    model log_endline = treatment;
```

```
    contrast "Parent Brochure Effect (Trt 2,4 vs Trt 1,3)"
             treatment -1 1 -1 1;
run;
quit;


/*****************************************************************

 * Question 1(c): Effect of Child Activity Booklet (Main Effect)
 * Contrast: (Trt 3,4) vs (Trt 1,2)
 *****************************************************************/


title1 "Contrast: Effect of Child Activity Booklet on Log-Endline
   Milk Consumption";
proc glm data=milk_log;
    class treatment;
    model log_endline = treatment;
    contrast "Child Activity Effect (Trt 3,4 vs Trt 1,2)"
             treatment -1 -1 1 1;
run;
quit;


/*****************************************************************

 * Question 2(a): Sex Differences in Fasting Blood Glucose
 * Test: Independent Two-Sample t-Test
 *****************************************************************/


title1 "Sex Differences in Fasting Blood Glucose";
proc ttest data=mydata.sugar;
    class sex;            /* 1 = Female, 2 = Male */
    var fasting;
run;
quit;

title1 "ANOVA for Fasting Blood Glucose by Sex";
proc glm data=mydata.sugar plots=diagnostics;
    class sex;
    model fasting = sex;
    means sex / hovtest=levene;
run;
quit;


```

```
/*************************************************************************
 * Question 2(b): Relationship Between Income and Fasting Blood
    Glucose
 * Test: One-Way ANOVA
 *************************************************************************/


title1 "ANOVA for Fasting Blood Glucose by Income Level";
proc glm data=mydata.sugar plots=diagnostics;
    class income;                    /* 1 = Lower, 2 = Middle, 3 =
       Higher */
    model fasting = income / solution;
    means income / hovtest=levene;
    lsmeans income / pdiff=all adjust=tukey;
run;
quit;


/*************************************************************************
 * Question 2(c): Intention-to-Treat Analysis
 * Model: One-Way ANOVA (Postmeal Blood Glucose)
 *************************************************************************/


title1 "ANOVA for Postmeal Blood Glucose by Treatment (Intention-to-
   Treat)";
proc glm data=mydata.sugar plots=diagnostics;
    class treatment;
    model postmeal = treatment / solution;
    means treatment / hovtest=levene;
    lsmeans treatment / pdiff=all adjust=tukey;
run;
quit;


/*************************************************************************
 * Question 2( d f ): Effect Modification by Sex (Treatment    Sex)
 * Model: Two-Way ANOVA
 *************************************************************************/


title1 "Two-Way ANOVA: Treatment    Sex Interaction for Postmeal
   Blood Glucose";
proc glm data=mydata.sugar;
```

```
    class treatment sex;
    model postmeal = treatment sex treatment*sex / solution;
    means treatment*sex / hovtest=levene;
run;
quit;

/* --- Interaction Plot --- */
title1 "Interaction Plot: Postmeal Blood Glucose by Treatment and
   Sex";
proc sgplot data=mydata.sugar;
    vline treatment / response=postmeal group=sex stat=mean markers
                      lineattrs=(thickness=2)
                      markerattrs=(symbol=circlefilled)
                      datalabel=sex;
    xaxis label="Treatment Group";
    yaxis label="Mean Postmeal Blood Glucose (mg/dL)";
    keylegend / title="Sex";
    title2 "Figure: Mean Postmeal Glucose by Treatment and Sex (
       Interaction Plot)";
run;
quit;


/*********************************************************************

 * Question 2(g): Effect Modification by Income (Treatment    Income
    )
 * Model: Two-Way ANOVA
 *********************************************************************/


title1 "Two-Way ANOVA: Treatment    Income Interaction for Postmeal
   Blood Glucose";
proc glm data=mydata.sugar plots=diagnostics;
    class treatment income;
    model postmeal = treatment income treatment*income / solution;
    means treatment*income / hovtest=levene;
run;
quit;

/* --- Interaction Plot --- */
title1 "Interaction Plot: Postmeal Blood Glucose by Treatment and
   Income";
proc sgplot data=mydata.sugar;
    vline treatment / response=postmeal group=income stat=mean
       markers
                      lineattrs=(thickness=2)
```

```
                        markerattrs=(symbol=circlefilled);
    xaxis label="Treatment Group";
    yaxis label="Mean Postmeal Blood Glucose (mg/dL)";
    keylegend / title="Income Level";
    title2 "Figure: Mean Postmeal Glucose by Treatment and Income (
       Interaction Plot)";
run;
quit;


/*********************************************************************

 * Question 2(h): Effect Modification by Fasting Blood Glucose
 * Model: ANCOVA (Treatment    Fasting Interaction)
 *********************************************************************/


title1 "ANCOVA: Treatment    Fasting Blood Glucose Interaction for
   Postmeal Response";
proc glm data=mydata.sugar plots=diagnostics;
    class treatment;
    model postmeal = treatment fasting treatment*fasting / solution;
run;
quit;

/* --- Publication-Quality Graph --- */
title1 "Interaction Plot: Postmeal vs. Fasting Blood Glucose by
   Treatment";
proc sgplot data=mydata.sugar;
    reg x=fasting y=postmeal / group=treatment;
    xaxis label="Fasting Blood Glucose (mg/dL)";
    yaxis label="Postmeal Blood Glucose (mg/dL)";
    keylegend / title="Treatment Group";
    title2 "Figure: Relationship between Fasting and Postmeal
       Glucose by Treatment";
run;
quit;
```

Listing 1: SAS Code for Homework 8

ANOVA for Log Transformed Andrea Milk Consumption

Two-Sample T-Test and CI

Sample	N	Mean	StDev	SE Mean
Male	30	4.1	0.0	0.0
Female	30	4.1	0.0	0.0

Difference = mu (Male) - mu (Female)
Estimate for difference: 0
95% CI for difference: (0, 0)
T-Test of difference = 0 (vs ≠): T-Value = *, P-Value = *, DF = 58

Two-Way ANOVA: Log Andrea Milk Consumption versus Sex, Activity Level

Method
Factor coding: (-1, 0, +1)

Factor	Type	Levels	Values
Sex	Fixed	2	Female, Male
Activity Level	Fixed	3	Active, Moderate, Sedentary

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Sex	1	0.000000	0.000000	*	*
Activity Level	2	0.000000	0.000000	*	*
Sex*Activity Level	2	0.000000	0.000000	*	*
Error	54	0.000000	0.000000		
Total	59	0.000000			

S	R-sq	R-sq(adj)	R-sq(pred)
0	100.00%	100.00%	100.00%

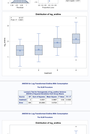
Term	Coef	SE Coef	95% CI	T-Value	P-Value	VIF
Constant	4.1	0	(4.1, 4.1)	*	*	
Sex						
Male	0	0	(0, 0)	*	*	1.00
Activity Level						
Active	0	0	(0, 0)	*	*	1.33
Moderate	0	0	(0, 0)	*	*	1.33
Sex*Activity Level						
Male Active	0	0	(0, 0)	*	*	1.33
Male Moderate	0	0	(0, 0)	*	*	1.33

Regression Equation
Log Andrea Milk Consumption = 4.1 + 0 Sex_Male + 0 Activity Level_Active + 0 Activity Level_Moderate + 0 Sex*Activity Level_Male Active + 0 Sex*Activity Level_Male Moderate

Fits and Diagnostics for Unusual Observations					
Obs	Log Andrea Milk Consumption	Fit	Resid	Std Resid	
1	4.1	4.1	0	*	X
2	4.1	4.1	0	*	X
3	4.1	4.1	0	*	X
4	4.1	4.1	0	*	X
5	4.1	4.1	0	*	X
6	4.1	4.1	0	*	X
7	4.1	4.1	0	*	X
8	4.1	4.1	0	*	X
9	4.1	4.1	0	*	X
10	4.1	4.1	0	*	X
11	4.1	4.1	0	*	X
12	4.1	4.1	0	*	X
13	4.1	4.1	0	*	X
14	4.1	4.1	0	*	X
15	4.1	4.1	0	*	X
16	4.1	4.1	0	*	X
17	4.1	4.1	0	*	X
18	4.1	4.1	0	*	X
19	4.1	4.1	0	*	X
20	4.1	4.1	0	*	X
21	4.1	4.1	0	*	X
22	4.1	4.1	0	*	X
23	4.1	4.1	0	*	X
24	4.1	4.1	0	*	X
25	4.1	4.1	0	*	X
26	4.1	4.1	0	*	X
27	4.1	4.1	0	*	X
28	4.1	4.1	0	*	X
29	4.1	4.1	0	*	X
30	4.1	4.1	0	*	X
31	4.1	4.1	0	*	X
32	4.1	4.1	0	*	X
33	4.1	4.1	0	*	X
34	4.1	4.1	0	*	X
35	4.1	4.1	0	*	X
36	4.1	4.1	0	*	X
37	4.1	4.1	0	*	X
38	4.1	4.1	0	*	X
39	4.1	4.1	0	*	X
40	4.1	4.1	0	*	X
41	4.1	4.1	0	*	X
42	4.1	4.1	0	*	X
43	4.1	4.1	0	*	X
44	4.1	4.1	0	*	X
45	4.1	4.1	0	*	X
46	4.1	4.1	0	*	X
47	4.1	4.1	0	*	X
48	4.1	4.1	0	*	X
49	4.1	4.1	0	*	X
50	4.1	4.1	0	*	X
51	4.1	4.1	0	*	X
52	4.1	4.1	0	*	X
53	4.1	4.1	0	*	X
54	4.1	4.1	0	*	X
55	4.1	4.1	0	*	X
56	4.1	4.1	0	*	X
57	4.1	4.1	0	*	X
58	4.1	4.1	0	*	X
59	4.1	4.1	0	*	X
60	4.1	4.1	0	*	X

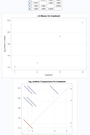
X denotes an observation whose X value gives it large influence.

Main Effects Plot for Log Andrea Milk Consumption



Interaction Plot for Log Andrea Milk Consumption

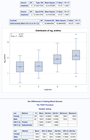
Cube Plot (Data Means) for Log Andrea Milk Consumption

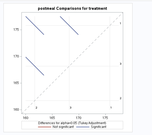


Pareto Chart of the Standardized Effects for Log Andrea Milk Consumption

Normal Plot of the Standardized Effects for Log Andrea Milk Consumption

Residuals Plots for Log Andrea Milk Consumption


Normal Probability Plot
Versus Fits
Histogram
Versus Order
Residual
Percent
Fitted Value
Frequency
Observation Order
0
0
0



Two-Way ANOVA: Treatment × Sex Interaction for Postmeal Blood Glucose

The GLM Procedure

HO: HbA1c

Class Level Information		
Class	Levels	Values
treatment	3	1 2 3
sex	2	Female Male

Number of Observations Read	300
Number of Observations Used	300

Two-Way ANOVA: Treatment × Sex Interaction for Postmeal Blood Glucose

The GLM Procedure

Dependent Variable: postmeal

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	14717.46597	2943.49319	32.58	<.0001
Error	294	26566.22075	91.95306		
Corrected Total	299	41013.68667			

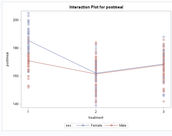
R-Square	Coeff Var	Root MSE	postmeal Mean
0.358869	5.874141	9.589710	163.3667

Source	DF	Type I SS	Mean Square	F Value	Pr > F
treatment	2	8691.148607	4345.574303	94.61	<.0001
sex	1	1798.369327	1798.369327	19.54	<.0001
treatment*sex	2	2625.948077	1312.974038	14.27	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
treatment	2	12810.38551	6337.044576	68.90	<.0001
sex	1	1878.14339	1878.14339	20.43	<.0001
treatment*sex	2	2625.30948	1456.654948	15.31	<.0001

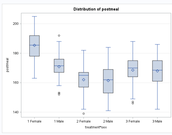
Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	167.9944905 B	1.17772440	142.66	<.0001
treatment 1	2.9950509 B	1.69530129	1.76	0.0784
treatment 2	-6.5272214 B	1.71367648	-3.81	0.0002
treatment 3	0.0000000 B	.	.	.
sex Female	-3.5743753 B	2.01431196	-1.77	0.0764
sex Male	0.0000000 B	.	.	.
treatment*sex 1 Female	13.0141406 B	2.85934536	4.47	<.0001
treatment*sex 1 Male	0.0000000 B	.	.	.
treatment*sex 2 Female	-0.2552083 B	2.85333598	-0.01	0.9893
treatment*sex 2 Male	0.0000000 B	.	.	.
treatment*sex 3 Female	0.0000000 B	.	.	.
treatment*sex 3 Male	0.0000000 B	.	.	.

Note: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.

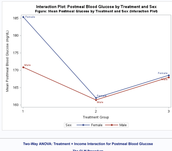


Two-Way ANOVA: Treatment × Sex Interaction for Postmeal Blood Glucose

The GLM Procedure



Least of treatment	Least of sex	postmeal	
		Mean	Std Dev
1	Female	169.3833333	10.1923629
1	Male	166.3883333	8.4139451
2	Female	161.4447552	9.4125401
2	Male	161.4576527	10.2556070
3	Female	168.1608634	8.9559542
3	Male	166.167	9.8864644
3	Male	166.167	9.8864644



Two-Way ANOVA: Treatment × Income Interaction for Postmeal Blood Glucose

The GLM Procedure

Class Level Information		
Class	Levels	Values
treatment	3	1 2 3
income	3	1 2 3

Number of Observations Read	300
Number of Observations Used	300

Two-Way ANOVA: Treatment × Income Interaction for Postmeal Blood Glucose

The GLM Procedure

Dependent Variable: postmeal

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	10484.52245	1306.06531	12.22	<.0001
Error	291	31146.14423	107.02113		
Corrected Total	299	41013.68667			

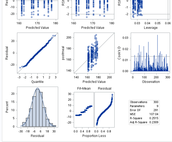
R-Square	Coeff Var	Root MSE	postmeal Mean
0.255668	6.107686	10.34610	169.3667

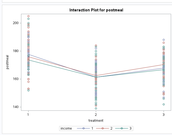
Source	DF	Type I SS	Mean Square	F Value	Pr > F
treatment	2	8691.148607	4345.574303	40.61	<.0001
income	2	256.360539	128.180269	1.20	0.3009
treatment*income	4	2137.013648	534.253412	4.99	0.0007

Source	DF	Type III SS	Mean Square	F Value	Pr > F
treatment	2	8745.350727	4372.675363	40.86	<.0001
income	2	278.427166	139.213584	1.30	0.2740
treatment*income	4	2137.013648	534.253412	4.99	0.0007

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	166.8157895 B	1.47635863	84.39	<.0001
treatment 1	4.9649123 B	2.09167466	2.71	0.0072
treatment 2	0.5131579 B	2.46782393	0.24	0.8356
treatment 3	0.0000000 B	.	.	.
income 1	2.9824561 B	2.45023295	1.90	0.0975
income 2	3.5175439 B	2.52694648	1.58	0.1650
income 3	0.0000000 B	.	.	.
treatment*income 1 1	2.5614115 B	3.63291709	0.85	0.4247
treatment*income 1 2	-8.5964912 B	3.58650543	-2.37	0.1126
treatment*income 1 3	0.0000000 B	.	.	.
treatment*income 2 1	-0.5073797 B	3.51538204	-0.28	0.7867
treatment*income 2 2	-0.3344461 B	3.51556000	-0.94	0.3249
treatment*income 2 3	0.0000000 B	.	.	.
treatment*income 3 1	0.0000000 B	.	.	.
treatment*income 3 2	0.0000000 B	.	.	.
treatment*income 3 3	0.0000000 B	.	.	.

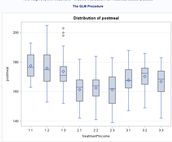
Note: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.





Two-Way ANOVA: Treatment × Income Interaction for Postmeal Blood Glucose

The GLM Procedure



Least of treatment	Least of income	postmeal	
		Mean	Std Dev
1	1	171.5937500	8.4440279
1	2	161.4230377	10.4733405
1	3	170.7241381	11.8290839
2	1	166.6666667	9.7982456
2	2	162.5161290	8.7761154
2	3	161.5535230	11.1729855
3	1	167.7612903	8.1112537
3	2	170.1333333	8.9370115
3	3	166.4137056	10.0948348

ANCOVA: Treatment + Fasting Blood Glucose Interaction for Postmeal Response

The GLM Procedure

Class Level Information		
Class	Levels	Values
treatment	3	1 2 3

Number of Observations Read	300
Number of Observations Used	300

ANCOVA: Treatment + Fasting Blood Glucose Interaction for Postmeal Response

The GLM Procedure

Dependent Variable: postmeal

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	11040.11004	2208.02201	21.24	<.0001
Error	294	30973.58653	103.98146		
Corrected Total	299	41013.68667			

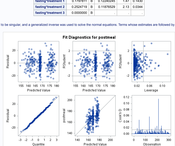
R-Square	Coeff Var	Root MSE	postmeal Mean
0.269213	6.244018	10.19713	169.3667

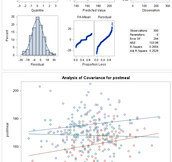
Source	DF	Type I SS	Mean Square	F Value	Pr > F
treatment	2	8691.148607	4345.574303	41.79	<.0001
fasting	1	559.518548	559.518548	5.38	0.0210
fasting*treatment	2	1789.442892	894.721446	8.61	0.0002

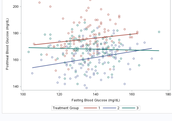
Source	DF	Type III SS	Mean Square	F Value	Pr > F
treatment	2	874.3004222	437.1502111	4.20	0.0160
fasting	1	539.2312775	539.2312775	5.19	0.0235
fasting*treatment	2	1789.442892	894.721446	8.61	0.0002

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Intercept	172.5435089 B	11.86223628	14.77	<.0001
fasting	-0.0276316 B	0.07541660	-0.37	0.7142
treatment 1	41.5453585 B	16.41453207	2.53	0.0119
treatment 2	0.0000000 B	.	.	.
fasting	-0.0302886 B	0.09941392	-0.31	0.7610
fasting*treatment 1	0.1758761 B	0.12340345	1.43	0.1540
fasting*treatment 2	0.2554719 B	0.11878525	2.15	0.0324
fasting*treatment 3	0.0000000 B	.	.	.

Note: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.







Design and Analysis of Randomized Trials in Public Health

Homework 8 Solutions

1. Refer to the dataset "milk".

A randomized trial was conducted to increase milk consumption in children 5-7 years old. Participating children were individually randomized to one of the following treatments. The interventions were delivered by the child's physician during an annual physical.

Treatment 1: Routine discussion with parent and child about healthy eating (standard of care)

Treatment 2: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children

Treatment 3: Standard of care + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Treatment 4: Standard of care + parents were given a brochure on healthy foods encouraging greater milk consumption by children + children were given an activity booklet with games and encouragement to eat healthy foods and especially to drink more milk

Note – this trial actually uses a factorial design, which we'll talk about soon.

Children's weekly milk consumption (measured in cups/week) was assessed during the annual physical ("baseline") and again three weeks later ("endline").

a. Conduct pairwise comparisons of the treatments at endline. How do the treatments differ (explain as if you were speaking to a general audience)? Which multiple comparisons method did you use, and why? Keep in mind from the last homework that you many have transformed your response before analysis – you need to apply that transformation going forward. (3 points)

Recall in the previous homework, we determined that we needed to do a natural log transformation of the endline variable. In that homework, we also found that children's weekly milk consumption differed following the interventions (the general F test was significant).

```
data milk; set milk;
        ln_endline=log(endline_milk);
run;

title1 'endline ANOVA - transformed';
proc glm data=milk plots=all;
        class treatment;
        model ln_endline = treatment;
        means treatment / hovtest=bf lines tukey;
run;
title1;
```

(LSMEANS also works in place of the MEANS statement)

The code above looks at all pairwise comparisons of the treatment means using a Tukey adjustment, which is a good method when you are interested in all pairwise comparisons. These comparisons are conducted on the transformed response variable.

The standard of care (treatment 1) and the treatment in which parents were given a brochure (treatment 2) did not result in differing milk consumption on average. However, an intervention in which children were given an activity book encouraging milk consumption did result in greater milk consumption on average, compared with these first two interventions, and an intervention in which both parents and children were provided with materials resulted in even higher milk consumption on average among young kids than the first three interventions.

Means with the same letter are not significantly different.

Tukey Grouping	Mean	N	treatment
A	2.16599	114	4
B	1.92872	75	3
C	1.48150	76	2
C			
C	1.41127	87	1

b. Overall, did a brochure given to parents affect children's milk consumption? (Note this is a question about the main effect of intervening with parents.) Write and test a contrast to answer this question using the endline data. Compare the two interventions that provided a brochure to the two interventions that did not. (3 points)

In the code above, adding the specific line below tests this question about the overall effect of giving parents a brochure. If there are no synergistic effects (i.e., no statistical interaction) between giving brochures and giving activity books, this comparison makes sense (it is interpretable). However, that "if" about synergistic effects is important, and it appears there is indeed a synergy in this dataset, i.e., (statistical) interaction between giving brochures and giving activity books. [We're getting to that in Topics covered since this homework.]

With the code below, it does appear that providing a brochure to parents has a significant effect on child milk consumption (p=0.0017).

```
estimate 'brochure' treatment -0.5 0.5 -0.5 0.5;
```

c. Overall, did an activity booklet given to children affect children's milk consumption? (Note this is a question about the main effect of intervening with children.) Write and test a contrast to answer this question using the endline data. Compare the two interventions that provided an activity booklet to the two interventions that did not. (3 points)

Adding the code below to your proc GLM, it appears that providing an activity book to children has a significant effect on child milk consumption ($p<0.0001$).

```
estimate 'activity book' treatment -0.5 -0.5 0.5 0.5;
```

d. Are your findings in questions 2 and 3 consistent with your findings in question 1? If yes, how did you assess consistency? If no, what seems to be the inconsistency? (2 points)

If we look at the Tukey groupings, the effect of the brochure in the absence of an activity book does not seem significant (that is, when we are comparing treatments 1 and 2). However, the effect of the brochure when kids were also given an activity book is significant (i.e., comparing treatments 3 and 4). This suggests that there is no "overall" effect of giving a brochure – its effect depends on whether you also give the activity book to kids. Similarly, the effect of giving an activity book to kids depends on whether or not parents received a brochure. [Statistical interaction in action!]

Means with the same letter are not significantly different.

Tukey Grouping	Mean	N	treatment
A	2.16599	114	4
B	1.92872	75	3
C	1.48150	76	2
C			
C	1.41127	87	1

2. Refer to the dataset "sugar".

Three hundred participants were enrolled into a study to evaluate the effect of three formulated meals on blood glucose. The participants were randomized to receive one of the three meals ("treatment"). All participants had their fasting blood glucose ("fasting") measured. They then consumed their assigned meal and their blood glucose was measured again after two hours ("postmeal"). During the study, data were collected on participant characteristics such as participant sex ("sex") and income. Income data were used to categorize participants into income tertiles ("income", where 1=lower income, 2=middle income, and 3=higher income). Units for blood glucose are mg/dL.

a. Were there sex differences in fasting blood glucose? Which statistical test did you use to answer this question? (1 point)

We can use an independent samples t-test or ANOVA to answer this question. Here is code for ANOVA:

```
proc glm data=sugar;
      class sex;
      model fasting=sex;
run;
```

There is insufficient evidence of sex differences in fasting blood glucose in this population (p=0.1814).

b. Was there a relationship between income and fasting blood glucose? How do you know? If there is a relationship, describe it. (1 point)

If we conduct an ANOVA (code below), we find evidence for differences in fasting blood glucose among the income groups (p<0.0001). If we do a pairwise comparison of the means for the three income groups, we find that the higher income group has significantly lower fasting blood glucose than the lower and middle income groups.

```
proc glm data=sugar;
      class income;
      model fasting=income;
      lsmeans income / lines adjust=tukey;
run;
```

c. Conduct an intention-to-treat analysis on the effect of the treatments on blood glucose (do not include any covariates). Use "postmeal" as your response variable. What do you conclude about the 3 test meals? Are your model assumptions satisfied? (1 point)

ANOVA indicates that blood glucose differs depending on which test meal (treatment) is consumed (p<0.0001). Pairwise comparison of the test meals indicates each test meal is significantly different from the others with respect to their effects on blood glucose. Specifically, test meal 1 results in the highest blood glucose, significantly higher than test meal 3, and test meal 2 results in the lowest blood glucose. Diagnostic plots and a formal test of constant variance across treatments indicates that model assumptions (normality, identical distribution, and constancy of error variance) appear reasonably satisfied.

```
proc glm data=sugar plots=all;
      class treatment;
      model postmeal=treatment;
      means treatment / hovtest=bf lines tukey;
run;
```

d. Does sex modify the effect of the treatments on blood glucose ("postmeal")? Make a graph to illustrate this, with clear axis labels and caption. This should be the sort of figure you'd put in a report or publication (you don't need to use SAS for graphing). (1 point)

Graphs may vary – they should clearly convey what's happening with the data, axes are labeled, etc. A bar graph would make the most sense, as both sex and the treatments are categorical variables. A reasonable caption would be something like: Average blood glucose among men and women after consuming one of three test meals.

e. Does sex modify the effect of the treatments on blood glucose ("postmeal")? To answer this question, write out your statistical model (not your estimated model – this should involve Greek letters with all symbols and subscripts explained). (2 points)

```
proc glm data=sugar;
      class treatment sex;
      model postmeal=treatment sex treatment*sex;
      lsmeans treatment*sex / slice=treatment;
run;
```

Corresponding statistical model:

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \epsilon_{ijk},$$

where *i*=1, 2, or 3,
j=1, 2,
k=1, 2, …, n_{ij} (the sample size for the *i*th treatment and *j*th sex),
y_{ijk} is the blood glucose of the *k*th person of sex *j* who consumes test meal *i*,
μ is the grand mean,
τ_i is the main effect on blood glucose of consuming test meal *i*,
β_j is the main effect on blood glucose of sex *j*,
$(\tau\beta)_{ij}$ is the interaction effect on blood glucose of test meal *i* and sex *j*,
and ϵ_{ijk} is the error term, with $\epsilon_{ijk} \sim iid\ N(0, \sigma^2)$, $\sigma^2 > 0$.

f. Does sex modify the effect of the treatments on blood glucose ("postmeal")? What do you conclude based on your analysis of the data? Explain as if you were presenting results to an audience of diabetes researchers. (2 points)

The graph and statistical test of treatment*sex interaction (p<0.0001) indicate that there are sex differences for the effect of treatment on blood glucose, but only for test meal 1 (see results of the slice statement, p<0.0001). (Of course, you'll want to phrase this differently for a technical but non-statistical audience.)

g. Does income modify the effect of the treatments on blood glucose ("postmeal")? Income tertile should be treated as a categorical variable. What do you conclude? (2 points)

No, there is insufficient evidence that income modifies the effect of the test meals on blood glucose (p=0.7342 for the treatment*income interaction), and, further, there is also insufficient evidence that income has any overall effect on blood glucose (p=0.2740 for the income main effect). Because the interaction effect was not significant, I did not do any further testing on the group means (e.g., didn't use lsmeans or the slice option).

```
proc glm data=sugar;
      class treatment income;
      model postmeal=treatment income treatment*income;
run;
```

h. Does fasting blood glucose modify the effect of the treatments on blood glucose ("postmeal")? Make a graph to illustrate this, with clear axis labels and caption. This should be the sort of figure you'd put in a report or publication. What do you conclude? (2 points)

No, there is insufficient evidence that fasting blood glucose modified the effects of the treatments (the test meals) on blood glucose two hours later (p=0.0955 for the treatment*fasting interaction).

In the graph you made, the lines may look non-parallel, which suggests some interaction, but your statistical test is not significant – perhaps there is indeed an interaction, but you have insufficient evidence in your sample to support that claim (at a significance level of 0.05 … if your significance level were 0.10, you'd conclude differently!).

```
proc glm data=sugar;
      class treatment;
      model postmeal=treatment fasting treatment*fasting / solution;
run;
```